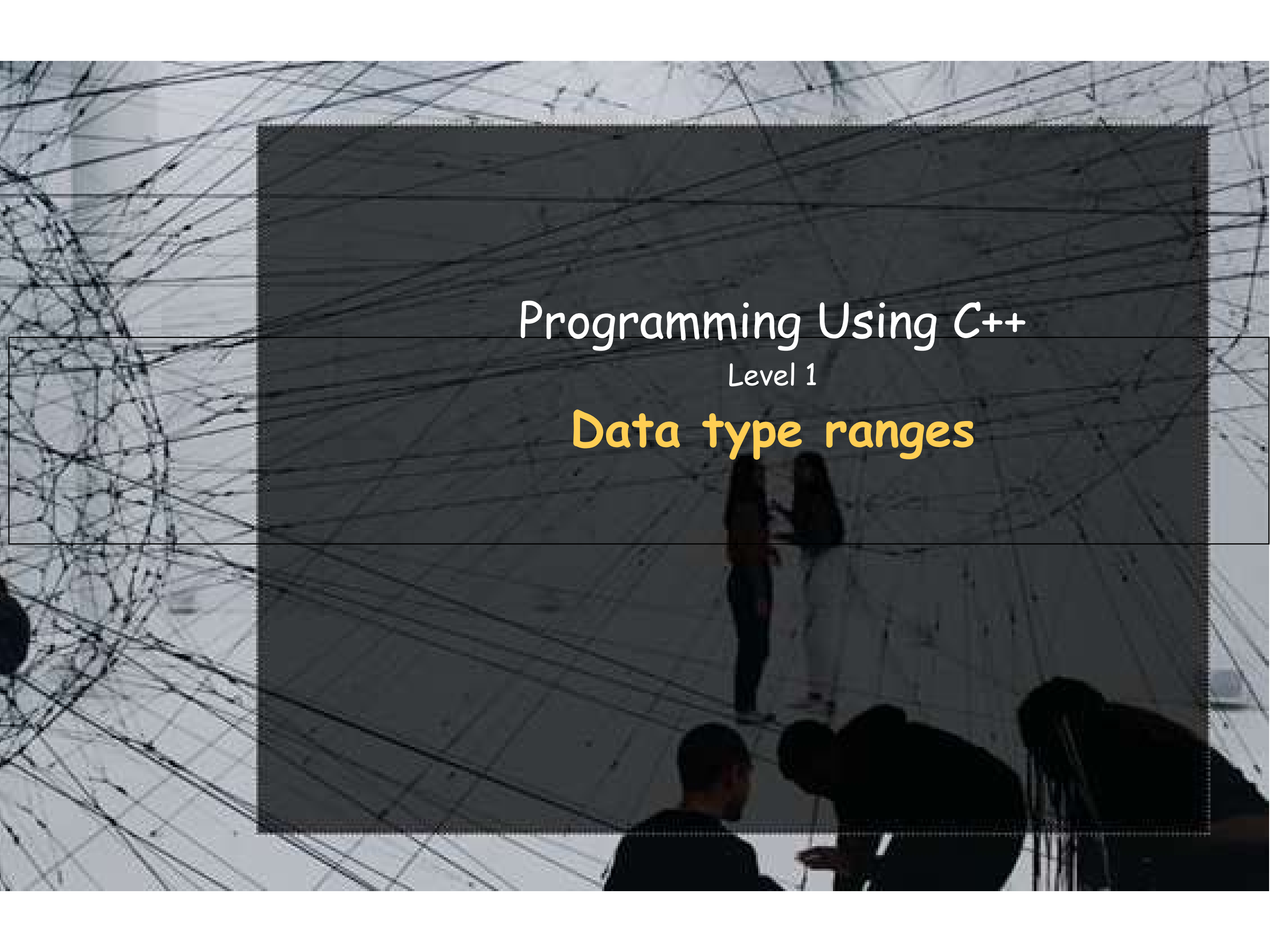


Programming Using C++ Level 1

Data type ranges



Programming Using C++

Level 1

Data type ranges

Definition

- ◆ Data type ranges define the minimum and maximum values that a variable of a specific data type can store.
- ◆ Different data types have different ranges depending on their size and whether they allow negative values.

Explanation

◆ In this example, the original:

`int Age;`

◆ was changed to:

`unsigned short Age;`

▣ `short` → uses less memory than a typical `int`

▣ `unsigned` → allows only non-negative values.

▣ `Age` → stores the user's age.

☐ Steps in main()

☐ Declare the variables

◆ Create the variables that will store the user's information.

☐ Use unsigned short for Age

◆ Store the age as a non-negative integer.

☐ Ask for the Name

◆ Use `getline()` to read the user's name.

☐ Ask for the Age

◆ Use `cin` to read the age.

☐ Use `cin.ignore()`

◆ Clear the leftover newline before using `getline()` again.

☐ Ask for the remaining information

◆ Read the city, country, salary, gender, and married status.

☐ Calculate Yearly Salary

◆ Multiply the monthly salary by 12

☐ Display the information

◆ Print all the values entered by the user.

☐ End the program

◆ Use `return 0;` to finish the program successfully.

💡 Example

```
//int Age;
```

```
unsigned short Age;
```

- ◆ int can store both negative and positive whole numbers.
- ◆ unsigned short` can store only non-negative whole numbers.

➡ Results

- ◆ If the user enters:

Please enter your Age:

43

- ◆ The value `43` is stored in the `Age` variable.

Note

- ◆ The exact range of a data type depends on the C++ implementation and system.
- ◆ For a commonly used implementation, unsigned short can store values from:
0 to 65,535
- ◆ short commonly stores values from:
-32,768 to 32,767

⊘ Warning

- ◆ unsigned types cannot represent negative values.
- ◆ If you try to store a negative value in an unsigned type, the result may not be what you expect.



Comparison

short

- ✓ Can store negative and positive whole numbers.
- ✓ Common range: -32,768 to 32,767
- ✓ Example: short Age;

unsigned short

- ✓ Stores only non-negative whole numbers.
- ✓ Common range: 0 to 65,535
- ✓ Example: unsigned short Age;

✦✦ Summary

- ✓ Data types have different ranges for the values they can store.
 - short → whole numbers, including negative values.
 - unsigned short → non-negative whole numbers.
 - unsigned → removes the negative part of the range.
 - The choice of data type should match the kind of values the variable needs to store.

Thank you 😊